A counterexample to plurisubharmonic variation of the Azukawa pseudometric

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Abstract

The bounded pseudoconvex family $D_t = \{(z,w) \in \mathbb{C}^2 : |z|^2 + |w|^2 + |w^2 + tz|^2 < 1\}$, $|t| < 1/4$, has smooth strongly convex fibers containing the origin. Nevertheless, the logarithmic Azukawa pseudometric in the fixed direction $(0,1)$ has Levi form $-\frac{1}{4}(1 - 1/\sqrt{5})$ at $t = 0$. This disproves the pseudometric assertion in Hosono's variation conjecture. The proof computes the Green function by a triangular polynomial automorphism and the Minkowski functional of a balanced domain. For this family, the reciprocal indicatrix volume equals the Bergman kernel at the origin, so the separate volume assertion still holds.

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1 Introduction

Hosono [2] proposed that the logarithmic Azukawa pseudometric should be plurisubharmonic under pseudoconvex variation of the underlying domains. He proved this assertion when the fibers are balanced about the fixed pole [2, Theorem 1.4 in the arXiv version]. We give an explicit counterexample with real-analytic strongly convex fibers. All Green functions and Azukawa pseudometrics in the example can be computed by completing a square.

Let $D \subset \mathbb{C}^n$ be a bounded hyperconvex domain containing the origin. We use the negative Green function with normalization

$$g_{D,0}(Z) = \sup\{u(Z) : u \in \mathrm{PSH}(D),\ u < 0,\ u(Z) \le \log\|Z\| + O(1) \text{ near } 0\}.$$

Following Hosono, we use the *logarithmic* Azukawa pseudometric

$$\mathcal{A}_{D,0}(X) = \limsup_{\lambda \to 0} \big(g_{D,0}(\lambda X) - \log|\lambda|\big). \tag{1.1}$$

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Thus $\mathcal{A}_{D,0}(cX) = \mathcal{A}_{D,0}(X) + \log|c|$ for $c \neq 0$, and $\mathcal{A}_{D,0}(0) = -\infty$. In this convention the Azukawa indicatrix is

$$I_{D,0} = \{X \in \mathbb{C}^n : \mathcal{A}_{D,0}(X) < 0\}.$$

For a pseudoconvex total space with hyperconvex fibers containing a common point, Hosono conjectured that $(t, X) \mapsto \mathcal{A}_{D_t,0}(X)$ is plurisubharmonic and that $t \mapsto -\log \mathrm{Vol}(I_{D_t,0})$ is subharmonic. These are, respectively, parts (1) and (2) of Conjecture 1.2 in the arXiv version of [2]. The following theorem disproves part (1).

THEOREM 1.1. *Let* $\Delta_{1/4} = \{t \in \mathbb{C} : |t| < 1/4\}$ *and set*

$$\mathcal{D} = \{(z, w, t) \in \mathbb{C}^2 \times \Delta_{1/4} : |z|^2 + |w|^2 + |w^2 + tz|^2 < 1\}, \tag{1.2}$$

$$D_t = \{(z, w) : (z, w, t) \in \mathcal{D}\}. \tag{1.3}$$

The total space $\mathcal{D}$ *is bounded and pseudoconvex. Every* D_t *is a bounded domain containing* 0*, with real-analytic strongly convex boundary, and is hyperconvex. Writing* $B = 1 + |t|^2$*, one has*

$$\mathcal{A}_{D_t,0}(0, 1) = \frac{1}{2} \log \left(\frac{1 + \sqrt{1 + 4/B}}{2} \right). \tag{1.4}$$

Consequently

$$\frac{\partial^2}{\partial t\, \partial \bar{t}} \mathcal{A}_{D_t,0}(0, 1)\Big|_{t=0} = -\frac{1}{\sqrt{5}(1 + \sqrt{5})} = -\frac{1}{4}\left(1 - \frac{1}{\sqrt{5}}\right) < 0. \tag{1.5}$$

Thus $(t, X) \mapsto \mathcal{A}_{D_t,0}(X)$ *is not plurisubharmonic.*

The example separates positivity in an individual tangent direction from positivity of the indicatrix volume. Indeed, in Section 4 we prove

$$K_{D_t}(0) = \frac{1}{\mathrm{Vol}(I_{D_t,0})},$$

where K_{D_t} denotes the diagonal Bergman kernel with respect to Euclidean volume. Berndtsson's theorem [1] therefore gives the volume assertion for this family. The same construction works in every fiber dimension at least two.

2 Geometry of the family

PROPOSITION 2.1. *The total space* $\mathcal{D}$ *in* (1.2) *is a bounded pseudoconvex domain. Its fibers are bounded hyperconvex domains with real-analytic strongly convex boundaries.*

Proof. Put

$$\rho_t(z, w) = |z|^2 + |w|^2 + |w^2 + tz|^2 - 1.$$

As a function of (z, w, t), it is plurisubharmonic, being a sum of squared moduli of holomorphic functions, minus a constant. Also $|z|, |w| < 1$ on $\mathcal{D}$. The function

$$-\log(-\rho_t(z, w)) - \log(1/16 - |t|^2)$$

is a plurisubharmonic exhaustion of $\mathcal{D}$: both logarithmic terms are bounded below there, and a sublevel set stays away from each boundary condition in (1.2). This proves pseudoconvexity once connectedness is verified below.

For each fixed t, completing the square gives

$$|z|^2 + |w^2 + tz|^2 = B\left|z + \frac{\bar{t}}{B} w^2\right|^2 + \frac{|w|^4}{B}. \tag{2.1}$$

Therefore the triangular polynomial automorphism

$$F_t(z,w) = \left(\sqrt{B}\, z + \frac{\bar{t}}{\sqrt{B}} w^2,\ w\right) \tag{2.2}$$

maps D_t biholomorphically onto

$$E_B = \{(u,v) \in \mathbb{C}^2 : |u|^2 + |v|^2 + |v|^4/B < 1\}. \tag{2.3}$$

Its inverse is $(u,v) \mapsto (u/\sqrt{B} - \bar{t}v^2/B, v)$. The domain E_B is convex and balanced, with a smooth real-analytic boundary. Every point of D_t is joined to 0 via $F_t^{-1}(sF_t(z,w))$, $0 \le s \le 1$, and the section $(0,0,t)$ connects the fibers. Thus $\mathcal{D}$ is connected. Smoothness of the fiber boundaries also follows from (2.2)–(2.3).

The function ρ_t is strictly plurisubharmonic in (z,w). It is a continuous negative exhaustion of the bounded fiber D_t, so the fibers are hyperconvex and strongly pseudoconvex. They are, in fact, strongly convex. To verify this stronger assertion, let $(a,b) \in \mathbb{C}^2$ be a real tangent direction and put $P = w^2 + tz$. The real Hessian satisfies

$$\begin{aligned}
\tfrac{1}{2}\operatorname{Hess}_{\mathbb{R}} \rho_t[(a,b)] &= |a|^2 + |b|^2 + |ta + 2wb|^2 + 2\Re(\bar{P}b^2) \\
&= B\left|a + \frac{2\bar{t}w}{B} b\right|^2 + \left(1 + \frac{4|w|^2}{B}\right)|b|^2 + 2\Re\big((\bar{w}^2 + \bar{t}\bar{z})b^2\big) \\
&\ge B\left|a + \frac{2\bar{t}w}{B} b\right|^2 + \left[1 - 2|t||z| + \left(\frac{4}{B} - 2\right)|w|^2\right]|b|^2.
\end{aligned}$$

For $|t| < 1/4$ and $|z| \le 1$, the bracket is at least $1/2$. The Hessian is consequently positive definite on the convex cylinder $\{|z| < 1\} \times \mathbb{C}$, which contains D_t. Its sublevel set D_t is convex, and the positive-definite Hessian on the boundary proves strong convexity. ∎

3 The Green function and the negative variation

We include the balanced-domain computation to make the proof independent of formulas for complex geodesics.

Lemma 3.1. *Let E be a bounded convex balanced domain containing 0, and let h be its Minkowski functional. Then $g_{E,0} = \log h$.*

Proof. The Minkowski functional is a complex norm. Its dual representation

$$h(Z) = \sup_{\ell \in E^\circ} |\ell(Z)|$$

shows that $\log h$ is plurisubharmonic. Here E° denotes the unit ball of the dual norm; the locally bounded supremum is continuous away from 0. Since E is bounded and contains a neighborhood of 0, h is comparable to the Euclidean norm. Thus $\log h$ is admissible in the definition of $g_{E,0}$, proving $g_{E,0} \ge \log h$.

Conversely, if u is any admissible function and $Z \in E \setminus \{0\}$, then $\zeta \mapsto u(\zeta Z/h(Z))$ is negative subharmonic on the unit disk with a logarithmic pole of weight at least one at 0. The subharmonic Schwarz lemma gives $u(\zeta Z/h(Z)) \le \log|\zeta|$. For completeness, this follows by applying the maximum principle to the removable extension of $u(\zeta Z/h(Z)) - \log|\zeta|$ on disks of radius $r < 1$, and then letting $r \uparrow 1$. Taking $\zeta = h(Z)$ gives $u(Z) \le \log h(Z)$. Supremizing over u proves the reverse inequality. ∎

PROPOSITION 3.2. *Let $B = 1 + |t|^2$, and let h_B be the Minkowski functional of E_B in (2.3). Then*

$$g_{D_t,0} = \log h_B \circ F_t, \qquad \mathcal{A}_{D_t,0}(X,Y) = \log h_B(\sqrt{B}\,X, Y).$$

More explicitly, for $(X,Y) \neq 0$,

$$\mathcal{A}_{D_t,0}(X,Y) = \frac{1}{2}\log\left(\frac{S_t + \sqrt{S_t^2 + 4|Y|^4/B}}{2}\right), \qquad S_t = B|X|^2 + |Y|^2. \tag{3.1}$$

In particular, the limsup in (1.1) is a limit.

Proof. For the domain E_B in (2.3), write h_B for its Minkowski functional. If $S = |u|^2 + |v|^2$, then h_B is determined by

$$\frac{S}{h_B^2} + \frac{|v|^4}{B h_B^4} = 1.$$

Solving for h_B^2 yields

$$h_B(u,v)^2 = \frac{S + \sqrt{S^2 + 4|v|^4/B}}{2}. \tag{3.2}$$

Biholomorphic invariance of the Green function and Lemma 3.1 give the complete formula

$$g_{D_t,0}(z,w) = \log h_B\left(\sqrt{B}\,z + \frac{\bar{t}}{\sqrt{B}}w^2, w\right). \tag{3.3}$$

For a fixed vector $(X,Y) \neq 0$, homogeneity gives

$$\begin{aligned} &g_{D_t,0}(\lambda X, \lambda Y) - \log|\lambda| \\ &\qquad = \log h_B\left(\sqrt{B}\,X + \lambda\frac{\bar{t}}{\sqrt{B}}Y^2, Y\right). \end{aligned}$$

Continuity of h_B proves both the existence of the limit in (1.1) and the stated formula. ∎

Proof of Theorem 1.1. Proposition 2.1 verifies the geometric assertions. Setting $X = 0$ and $Y = 1$ in (3.1) gives (1.4). Differentiating $q(B) = \frac{1}{2}\log((1 + \sqrt{1 + 4/B})/2)$ gives

$$q'(B) = -\frac{1}{B^2\sqrt{1 + 4/B}\,(1 + \sqrt{1 + 4/B})}.$$

Since $B = 1 + |t|^2$, the Levi form at $t = 0$ is $q'(1)$, which is (1.5). ∎

The failure can also be seen without differentiation. The function in (1.4) depends only on $|t|$ and is strictly decreasing as $|t|$ increases. For every $0 < r < 1/4$ it therefore satisfies the strict reverse submean inequality

$$\mathcal{A}_{D_0,0}(0,1) > \frac{1}{2\pi}\int_0^{2\pi} \mathcal{A}_{D_{re^{i\theta}},0}(0,1)\,d\theta.$$

REMARK 3.3. *The distinguished direction also has a direct geometric description. Let $r_t > 0$ be the solution of $r_t^2 + r_t^4/B = 1$. By (2.1), the projection $(z,w) \mapsto w$ maps D_t onto the disk Δ_{r_t} and has a holomorphic right inverse*

$$\sigma_t(w) = \left(-\frac{\bar{t}}{B}w^2, w\right), \qquad w \in \Delta_{r_t}.$$

Moreover, $\sigma_t(0) = 0$, $d\sigma_t(0)(1) = (0,1)$, *and* (3.3) *gives* $g_{D_t,0}(\sigma_t(w)) = \log(|w|/r_t)$. *Consequently* $\mathcal{A}_{D_t,0}(0,1) = -\log r_t$, *as also follows from Proposition 3.2. The radius* r_t *increases with* $|t|$, *which explains the negative variation.*

The maps F_t *and* σ_t *are holomorphic in the fiber variables for each fixed* t; *their dependence on* t *is not holomorphic. Thus the reduction to balanced fibers is a fiberwise computation and does not put the total family under Hosono's balanced-domain theorem.*

4 The indicatrix-volume assertion

PROPOSITION 4.1. *For the family in Theorem 1.1, the negative logarithm of the Azukawa indicatrix volume is subharmonic. More precisely, writing* $I_t = I_{D_t,0}$ *and*

$$s(B) = \frac{\sqrt{B^2+4B} - B}{2},$$

one has

$$\mathrm{Vol}(I_t) = \frac{\pi^2 s(B)(4 - s(B))}{6B}, \qquad K_{D_t}(0) = \frac{1}{\mathrm{Vol}(I_t)}.$$

Proof. Formula (3.1) gives the actual Azukawa indicatrix

$$I_t = \{(X,Y) : B|X|^2 + |Y|^2 + |Y|^4/B < 1\}. \tag{4.1}$$

Since $s = s(B)$ satisfies $s + s^2/B = 1$, integrating first in X, then setting $u = |Y|^2$, yields

$$\begin{aligned}\mathrm{Vol}(I_t) &= \frac{\pi^2}{B}\int_0^{s(B)} (1 - u - u^2/B)\, du \\ &= \frac{\pi^2}{B}\left(s - \frac{s^2}{2} - \frac{s^3}{3B}\right) = \frac{\pi^2 s(4-s)}{6B}.\end{aligned} \tag{4.2}$$

The Bergman kernel of a bounded balanced domain at its center equals the reciprocal of its volume: averaging a normalized holomorphic function under scalar rotations gives the constant function, which minimizes the L^2 norm. Using F_t and $|\det dF_t(0)|^2 = B$, we obtain

$$K_{D_t}(0) = \frac{B}{\mathrm{Vol}(E_B)} = \frac{1}{\mathrm{Vol}(I_t)}.$$

Berndtsson's variation theorem [1] therefore implies that $-\log \mathrm{Vol}(I_t)$ is subharmonic. ∎

COROLLARY 4.2. *For every* $n \geq 2$, *there is a bounded pseudoconvex domain in* $\mathbb{C}^n \times \Delta_{1/4}$ *whose fibers have real-analytic strongly convex boundaries and contain the origin, but whose logarithmic Azukawa pseudometric is not plurisubharmonic in the parameter and the tangent vector.*

Proof. Use the fibers

$$D_t^{(n)} = \left\{(z,w,\zeta) \in \mathbb{C}^2 \times \mathbb{C}^{n-2} : |z|^2 + |w|^2 + \|\zeta\|^2 + |w^2 + tz|^2 < 1\right\}.$$

The exhaustion and Hessian arguments of Proposition 2.1 apply after adding $\|\zeta\|^2$. Extend F_t by the identity on ζ. Its image is the convex balanced domain

$$\left\{(u,v,\zeta) : |u|^2 + |v|^2 + \|\zeta\|^2 + \frac{|v|^4}{B} < 1\right\}.$$

Lemma 3.1 then gives the same value (1.4) in the fixed direction $(0,1,0)$, and hence the same negative Levi form (1.5). ∎

AI declaration and verification status

The research question and proposed strategy were supplied by Shijie Bao. OpenAI Codex assisted with the construction, calculations, and exposition. This version has not undergone independent mathematical review.

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